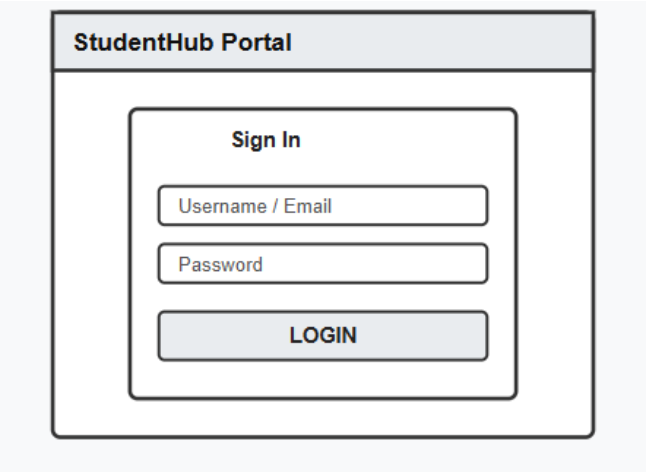
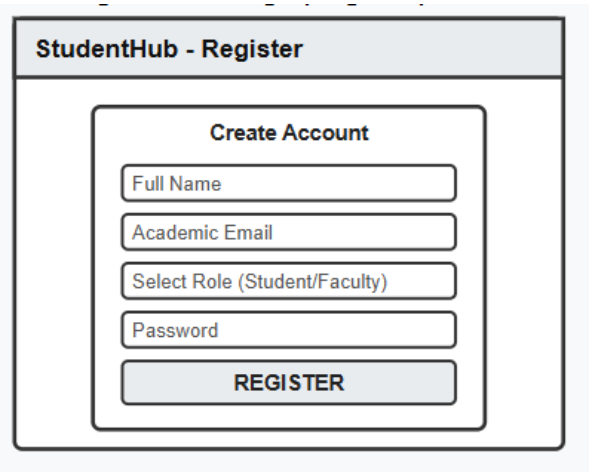
Aim: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

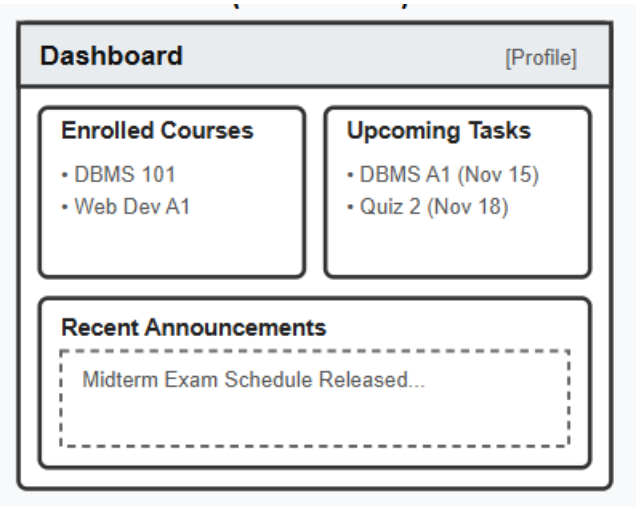
Tools/Technology to be Used: VS Code, Draw.io/Figma, Git, GitHub

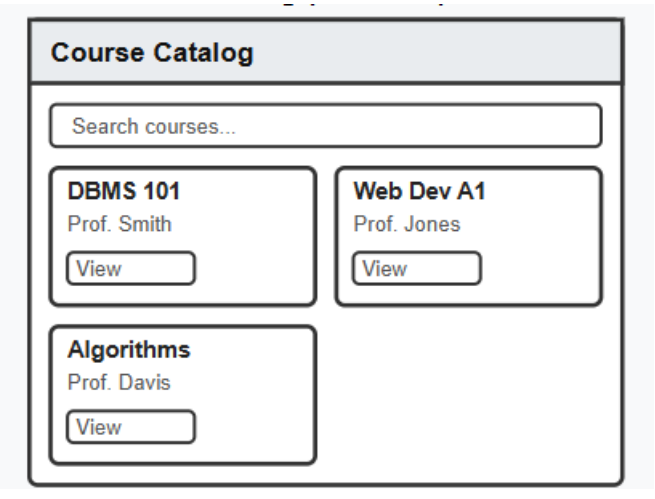
Theory Logic: Requirement analysis, sitemap design, wireframing, Git/GitHub, README writing, folder structuring.

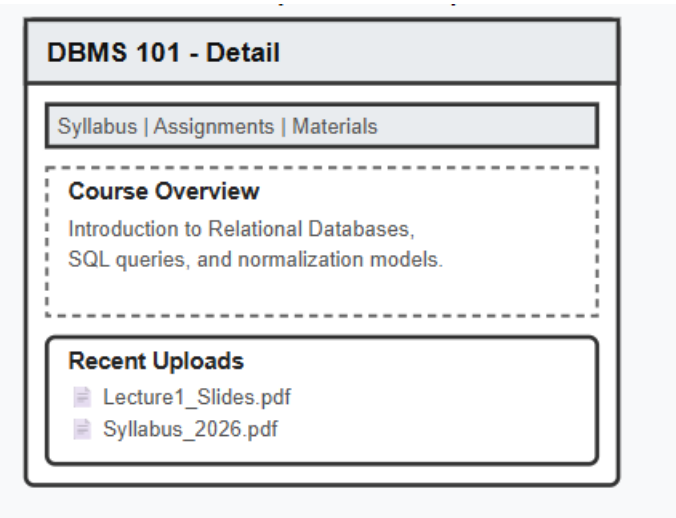
Wireframe:











Question-Answer:

1. What is URL and parts of URL? Protocol: (https://) Method used to fetch data.

   A URL (Uniform Resource Locator) is the web address used to find resources on the internet.

- Domain: (example.com) Host server name.
- Port: (:443) Communication endpoint.
- Path: (/courses/index.html) File location on the server.
- Query: (?id=101) Extra data sent as key-value pairs.
- Fragment: (#section2) Specific section on the page.

2. How does HTML file processed in web browser? Fetch: Browser downloads the raw HTML bytes.

1. Parse: Converts bytes to tokens and builds the DOM Tree (structure).
2. Style: Parses CSS to build the CSSOM Tree (styling).
3. Render: Combines DOM and CSSOM into a Render Tree.
4. Layout & Paint: Calculates exact element positions and draws pixels on the screen.

3. How will page navigation flow managed among all HTML pages?

- **Relative Anchor Links**: Connect HTML pages using <a href="page.html">.
- **Shared Navigation Bar**: Place the same header menu across all HTML files.

- **URL Parameters**: Pass IDs in links (e.g., course-detail.html?id=dbms) to load dynamic content.
- **Folder Structure**: Organize pages systematically into directories (e.g., pages/admin/).

4. How will GitHub commits be maintained after each practical?

- Check changes: git status
- Stage files: git add .
- Commit with message: git commit -m "Practical X: [Brief description of work done]"
- Push to GitHub: git push origin main

Conclusion:

By establishing the problem scope, core user roles, and module navigation upfront, the StudentHub project creates a structured operational roadmap for the semester. Defining the sitemap and low-fidelity wireframes early guarantees consistent UI and layout patterns across all 10 pages, while the modular directory structure and Git workflow ensure seamless scalability and maintenance as additional backend functionality is introduced.